

Knowledge Representation in AI

Semantic Net Frame

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Semantic Networks

- Weak Slot-and-Filler structure
- Non formal K. representation (Predicate Logic is formal)
- Introduced by Quillian, 1968.

Underlying idea:

Meaning of a concept comes from the way in which it is associated with other concepts.

“A semantic net is a graphical representation of relations between elements/objects in a domain, where domain elements / pieces of info. are represented by a set of interconnected “Nodes” & the relations between them are represented by a set of “labelled arcs/links”.

- A **semantic net** is a graph-based structure used to represent knowledge in the field of artificial intelligence (AI). It is comprised of **nodes** representing concepts or entities and **edges** that indicate the relationships between them. The structure is often used to illustrate how different concepts are connected in a meaningful way.

“Horses eat grass”



-A link asserts something to be true about one element relative to another.

-A link is fundamentally a “**Binary relation**”

Some commonly used binary relations are:

Is a instance

Member-of Has-parts

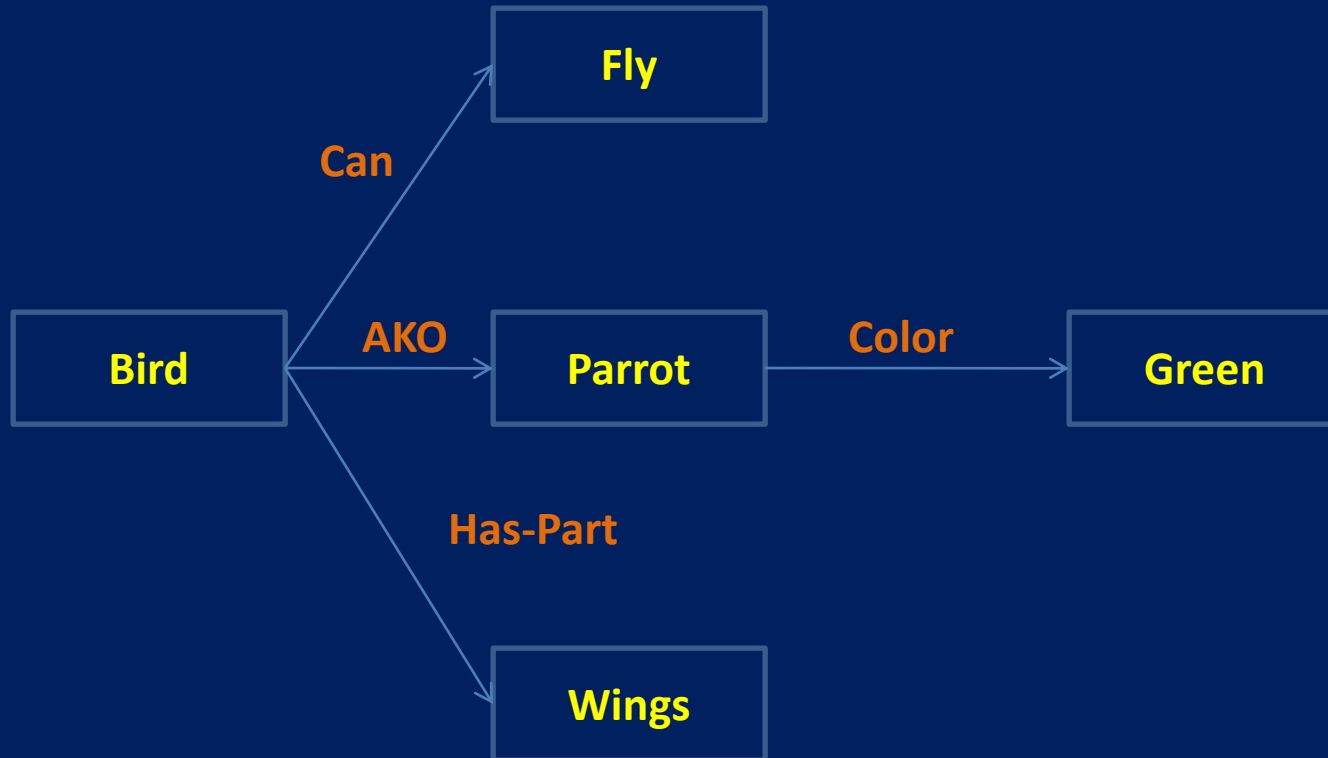
Subset-of Agent

A-kind-of(AKO) Color

Part-of

And other domain specific relations

Representing Binary Predicates



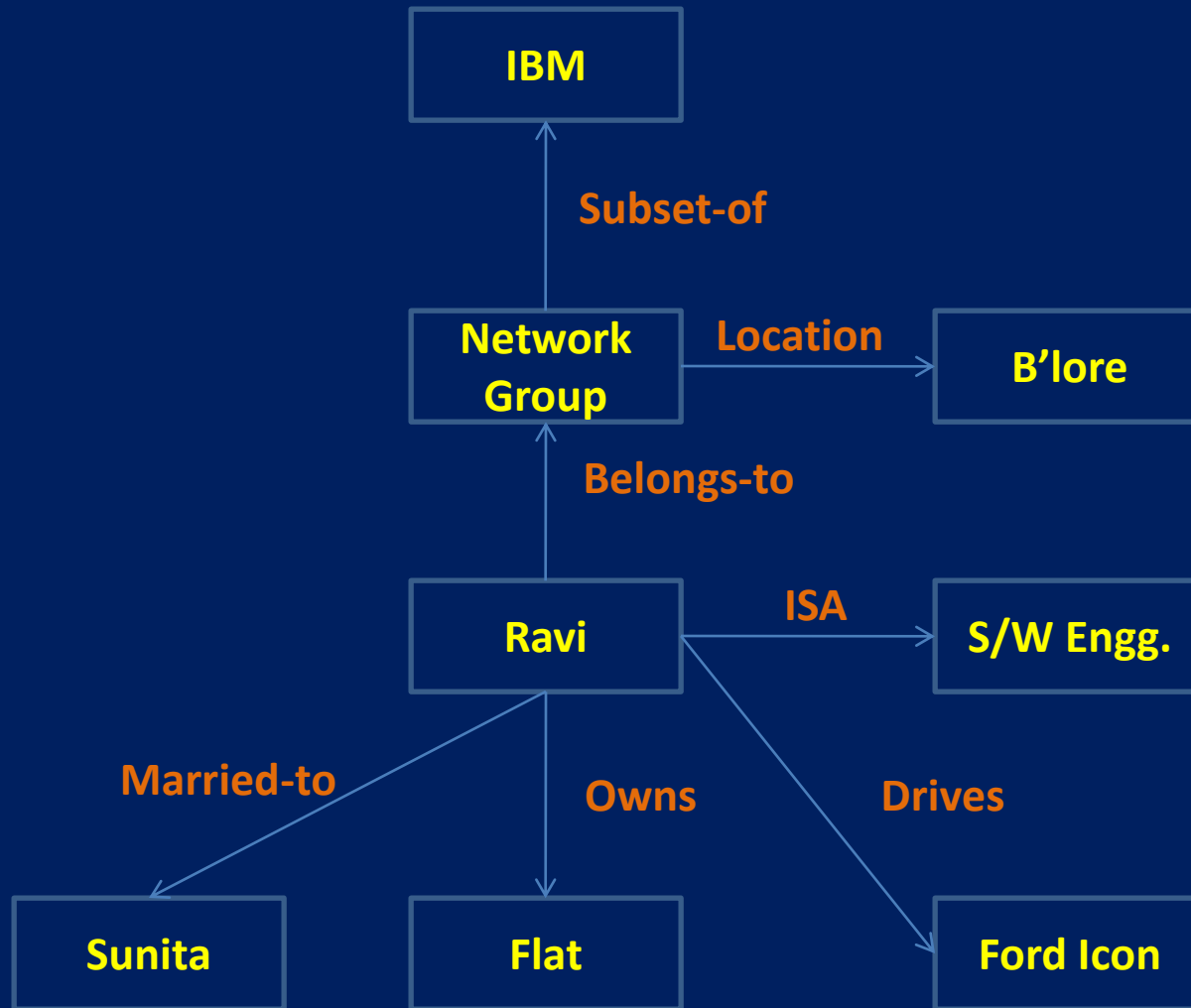
Predicates:

Color(parrot, green)

can(bird, fly)

AKO(parrot, bird)

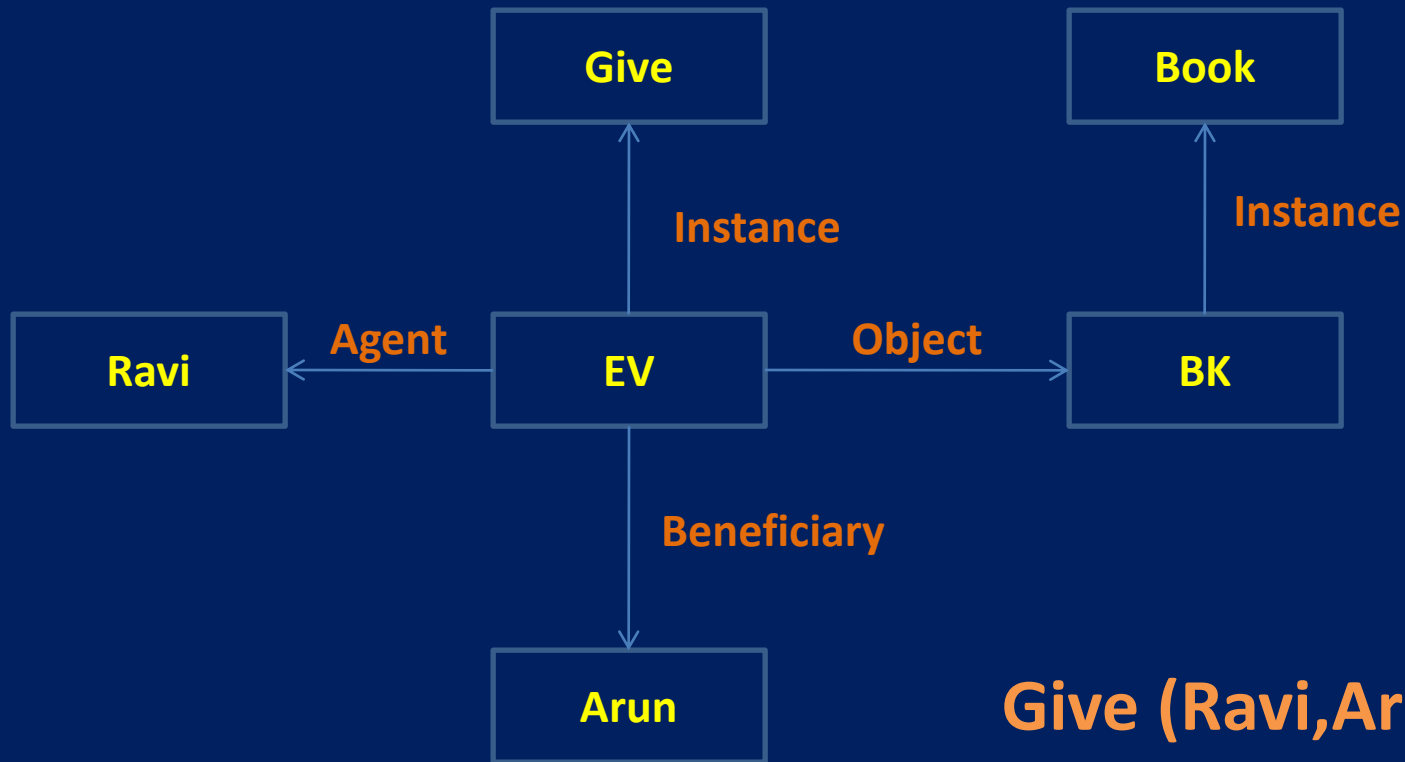
has-part(bird, wings)



Representing Non-Binary Predicates

Convert to binary by creating a new object representing the predicate.

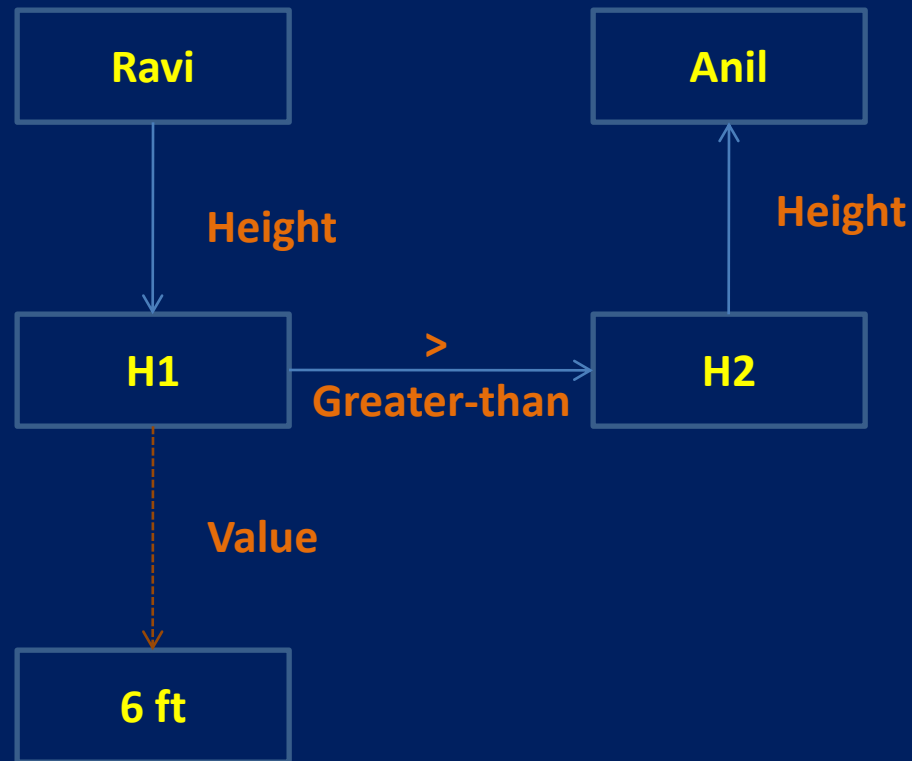
“Ravi gave the book to Arun”



Give (Ravi,Arun,book)

Typical Relations

“Ravi is taller than Anil”



Property Inheritance

(Inferencing in Semantic nets)

Any property that one asserts to be true for a class of elements should also be true for any instance of the class.

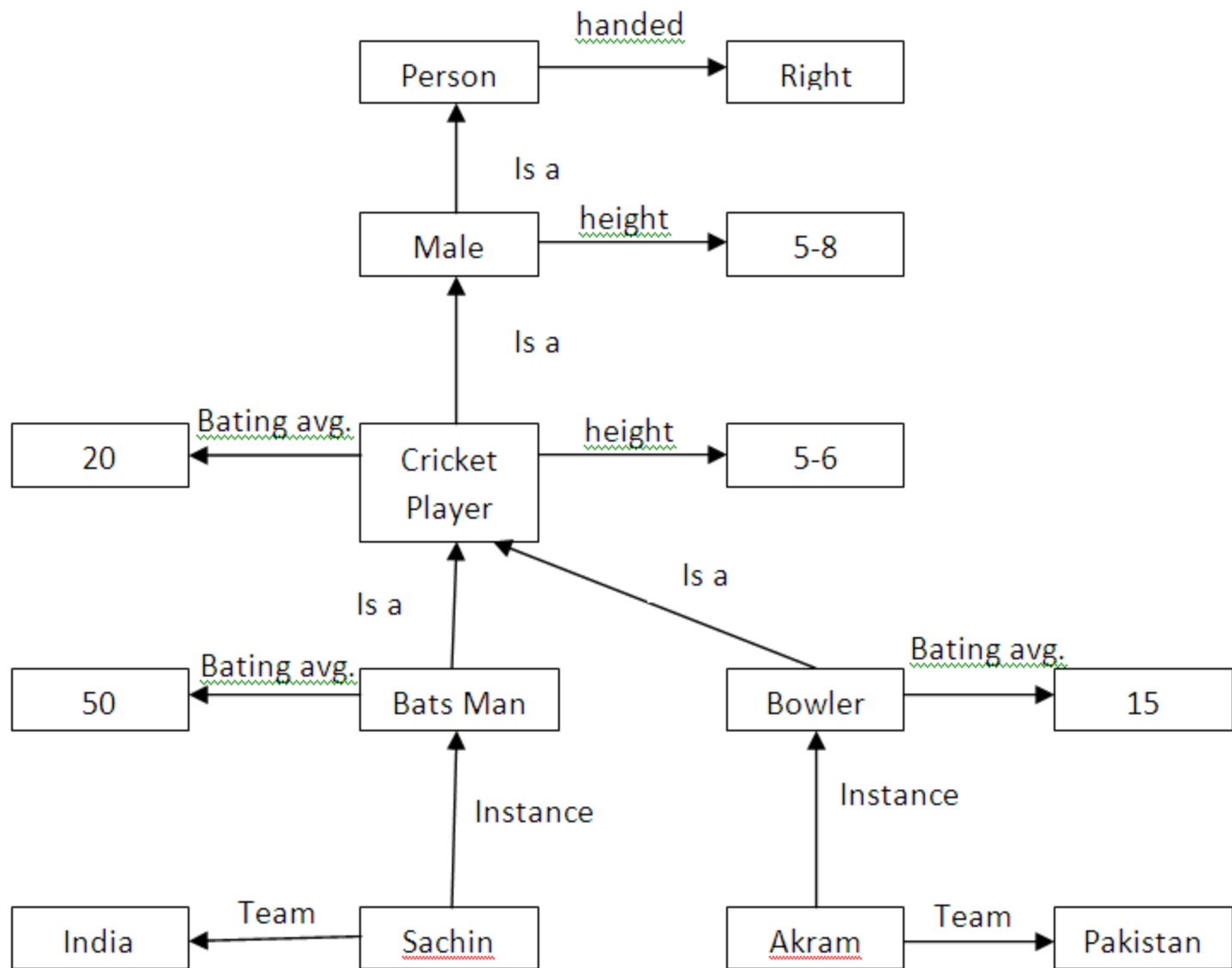
If a class 'X' has some property P and if 'x' is a member/instance of 'X'

THEN one can infer that 'x' has property P.

Algorithm

To retrieve a value for an attribute of object **O** which is an instance of class **C**.

1. Locate **O** in the KB represented as semantic net.
2. If a value for the desired attribute is directly available then Return the value
 - A. Else, check if there is a value for the attribute instance, if not then return failure
 - B. Else, move to the next node corresponding to that value (when a range is given) & find a value for the attribute, if found return it.
 - C. Else, continue until there is no value for the **is-a attribute** or until answer is found
 - i. Get a value of the **is-a attribute** & move to that mode.
 - ii. See, if there is a value for the attribute if yes, then return it.



Team (Sachin) = India
Height (Akram) = 5-6

Batting-avg (Sachin) = 50
Sachin-age = Not found

Frames

- Weak slot-and-filler structure

Basis:

- Human beings have the ability to interpret new situations on the basis of knowledge gained from experience in similar situations.

Definition:

A frame is a collection of attributes (slots) & associated values which describe some entity. A collection of frames related to each other form a “frame system” used to encode knowledge & support reasoning.

- A slot may contain
 1. Range of values
 2. Default values
 3. Conditions
 4. Pointers to other related slots
 5. Procedures that are activated when needed
 6. Subfields (facets)

Frame: **CAR**

Specialization of : land vehicle

Model:

Range: (family model, convertible, 2-door)

Default: family model

Body: steel

Windows: glass

Mobility mechanism: has wheels

Fuel:

range: (petrol, gas, diesel)

default: petrol

Fuel remaining:

range: (empty, half, full,...)

default: none

If-Needed: Check fuel meter //adding a procedure

[Contains such attributes that represent our expectation of a car]

Reasoning with Frames

In situations when incomplete information is not available then a suitable frame can be used to infer facts which are not explicitly observed.

e.g., if an object is identified as a car, one can infer that it should have a steering wheel.

Reasoning process

1. Select a frame to represent the current situation (i.e., choose a frame which best fits to the current context based on the available partial evidence/observation)
2. Instantiate the selected frame on the basis of specific current conditions
3. Infer values for missing slots or trigger procedures

Drawback:

Can one call a “toy car” as a “car”?